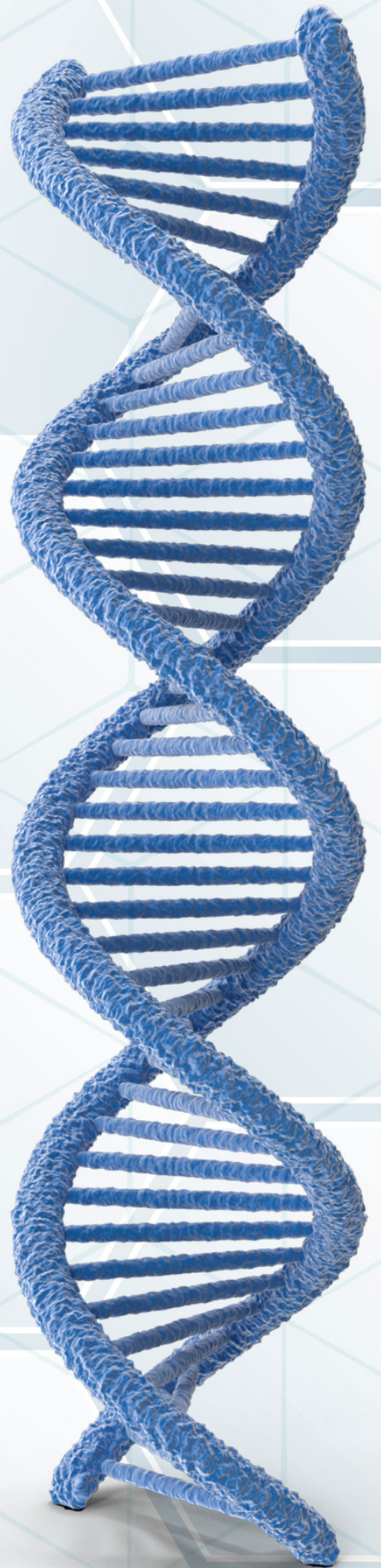


Bibliografia Internazionale sul Collagene



1st Abstract

Specific Collagen Peptides Improve Bone Elastic Structure in Postmenopausal Women

Daniele Zeggio*

* Medical Doctor specialist in Neurology and Nutritionist. Independent Researcher started his researches about collagen supply bones related in 1992. 36100 Vicenza, Italy

* Correspondence: zeggio.d@libero.it; Tel: +39 335 5361830

Published: 20 December 2020

Abstract introduction: Investigations in rodents as well as in vitro experiments have suggested an anabolic influence of specific collagen peptides (SCP) on bone quality both bone formation and bone mineral density (BMD). The goal of the study was to investigate the effect of 120 days daily oral administration of 15 g SPC group on BMD in postmenopausal women with primary, osteoporosis reduction of BMD undergoing drug holiday and food diary control. Methods: 10 women were enrolled in this study denominated “area test 120”. The primary endpoint was the change in BMD after 120 days on either the left femoral neck or the lumbar spine. Results: A total of 9 woman completed the study, but all subjects were included in the intention-to-treat (ITT) analysis (age 64.3 7.2 years; Body Mass Index, BMI 23.6 3.6 kg/m²; average T-score -2,61). In the SCP group the BMD increased significantly (average T-score -2,13 p=0.184).

Conclusion: These data demonstrate that the intake of SCP increased BMD in postmenopausal women with primary, osteoporosis reduction of BMD undergoing drug holiday and diary food control. In addition, SCP supplementation was associated with a favourable shift in bone markers, indicating increased bone formation and reduced bone degradation.

Publicazione al WCO LONDRA (Congresso Mondiale Osteoporosi e patologie osteoarticolari 2021)

2nd Abstract

Effects of Goose Collagen and Collagen Peptide on Osteoporosis

Pingping Wang, Jinli Jang, DAodang Pan, Yangying sun, Zhen Wu, Jinxuan cao

Introduction:

The effects of collagen and collagen peptide from goose bone on osteoporosis were studied. Fourier transform infrared spectrometry, ultraviolet spectrum, differential scanning calorimetry, and sodium dodecyl sulfate polyacrylamide gel electrophoresis identified the goose collagen as typical type I collagen. Results showed that osteopontin, bone morphogenetic protein 2, runt-related transcription factor 2, and type I collagen levels in bone marrow mesenchymal stem cells supernatant were dramatically higher in the experimental groups than in the control group ($p < 0.05$), and was highest in the experimental group composed of low molecular weight collagen peptide linked with calcium. Then the osterix, runt-related transcription factor 2, transforming growth factor- $\beta 1$, and bone morphogenetic protein 2 expression levels related to osteogenic differentiation were detected by real-time fluorescent quantitative polymerase chain reaction. The findings showed that goose collagen and collagen peptide can improve the gene expression related to osteogenic differentiation which reflected the level of bone marrow mesenchymal stem cells to osteoblasts.

Conclusions:

Collagen and collagen peptide were successfully extracted from goose bone with pepsin and trypsin under different conditions. FT-IR and UV investigations showed that the extracted substances were typical Col I and collagen peptide. SDS-PAGE and SEM showed that the molecular weight of the collagen and collagen peptide differed, although the triple-helical structure was intact. There was also a difference in thermal stability. ELISA investigations indicated that OPN, BMP-2, Col I, and Runx2 expression all increased to varying degrees when collagen and collagen peptides were administered. The results of RT-PCR on day 14 showed that BMP-2, OSX, Runx2, and TGF- $\beta 1$ secretion all increased significantly. Therefore, the effect of collagen and collagen peptide on OP is well supported.

3rd Abstract

Role of collagen hydrolysate in bone and joint disease

Roland W. Moskowitz, Case Western Reserve University
division of Rheumatic Diseases

Objectives: To review the current status of collagen hydrolysate in the treatment of osteoarthritis and osteoporosis.

Methods: Review of past and current literature relative to collagen hydrolysate metabolism, and assessment of clinical investigations of therapeutic trials in osteoarthritis and osteoporosis.

Results: Hydrolyzed gelatin products have long been used in pharmaceuticals and foods; these products are generally recognized as safe food products by regulatory agencies. Pharmaceutical-grade collagen hydrolysate (PCH) is obtained by hydrolysis of pharmaceutical gelatin. Clinical studies suggest that the ingestion of 10 g PCH daily reduces pain in patients with osteoarthritis of the knee or hip; blood concentration of hydroxyproline is increased. Clinical use is associated with minimal adverse effects, mainly gastrointestinal, characterized by fullness or unpleasant taste. In a multicenter, randomized, double-blind, placebo-controlled trial performed in clinics in the United States, United Kingdom, and Germany, results showed no statistically significant differences for the total study group (all sites) for differences of mean pain score for pain. There was, however, a significant treatment advantage of PCH over placebo in German sites. In addition, increased efficacy for PCH as compared to placebo was observed in the overall study population amongst patients with more severe symptomatology at study onset. Preferential accumulation of ¹⁴C-labeled gelatin hydrolysate in cartilage as compared with administration of ¹⁴C-labeled proline has been reported. This preferential uptake by cartilage suggests that PCH may have a salutary effect on cartilage metabolism. Given the important role for collagen in bone structure, the effect of PCH on bone metabolism in osteoporotic persons has been evaluated. Studies of the effects of calcitonin with and without a collagen hydrolysate-rich diet suggested that calcitonin plus PCH had a greater effect in inhibiting bone collagen breakdown than calcitonin alone, as characterized by a fall in levels of urinary pyridinoline cross-links. PCH appeared to have an additive effect relative to use of calcitonin alone.

Conclusions: Collagen hydrolysate is of interest as a therapeutic agent of potential utility in the treatment of osteoarthritis and osteoporosis. Its high level of safety makes it attractive as an agent for long-term use in these chronic disorders.

4th Abstract

Response to Hydrolyzed Collagen Protein Supplementation in a Cohort of Pregnant and Postpartum Women

Leon Baginski, Marc Winter, Thomas S Bailey, Scott Capobianco⁴, Marsha Granese, Freddie Granger MA, Kurt Miller, Kerry Price, Sara Ramirez, Craig Salcido, Frank Turner and Mary O'Toole¹²

Abstract: There is overwhelming evidence that links maternal nutrition during pregnancy with fetal outcomes. This study evaluated the safety and efficacy of Protiva treatment responses in pre and postpartum female subjects between the ages of 19 years and 43 years who entered into this open label study. The protocol was designed to take a twice daily collagen protein supplementation drink during the third trimester and for 10 weeks postpartum. The outcomes were measured against a control group of patients who did not take the protein supplementation. The differences between mean score for the physical health domain of WHOQOL-BREF scale in the Protiva group and the control group were different and met statistical significance ($p=0.0003$). The study found that 100% of control patients and 95% of Protiva patients had initial serum protein levels below the normal median range and 27% of control patients and 33% of Protiva patients were below normal range. With collagen protein supplementation we were able to demonstrate improved quality of life and wound healing rates and 100% improvement in the protein levels for Protiva patients and a 72% improvement over control patients. Protiva Pregnancy and Protiva New Mom was determined to be safe and well tolerated when taken during pregnancy and postpartum during the study.

Conclusion: Pregnancy and lactation place increased nutritional demands on the pregnant woman. The need for protein also increases, for proper fetal and placental growth and development. The average non-pregnant woman needs approximately 46 g of protein daily and this increases to 71-76 (g/dL) in pregnancy. This study found the great majority of patients had initial serum levels below the normal median and even completely below the normal range. With collagen protein supplementation we were able to demonstrate improved quality of life and wound healing and 100% improvement in protein levels in study patients and a 72 improvement over control patients.

5th.Abstract

Postmenopausal osteoporosis. Treatment with calcitonin and a diet rich in collagen proteins

M. Adam; P. Spacek; h. Hulejová, j. Blahos

Background: Calcitonin has an important role in the treatment of postmenopausal osteoporosis. The authors investigated the effect of calcitonin administration or calcitonin administration supplement with a diet rich in collagen proteins on markers of bone metabolism.

Methods and results: A group of 108 patients with postmenopausal osteoporosis (BMD lower than 80% of the BMD in premenopausal women) was treated with Calsynar (Rhône-Poulenc-Rorer), 100 u., i.m. twice a week for 24 weeks. Forty-nine of these women took an oral collagen hydrolysate, 10 g per day, for the same period of time. Before and after termination of treatment clinical and laboratory tests were made, X-ray examination of the LS spine and the right forearm, single-photon osteometry of the right forearm and urinary excretion of pyridinoline (UPD), deoxypyridinoline (UDPD) and hydroxyproline (Uhyp) was assessed. As a result of treatment the BMD values increased only insignificantly (by 1.8%) the UPD values declined (to 62.51%) and those of UDPD (to 70.4%), as compared with basal values. The statistical significance is at the 1% level. When collagen proteins were administered concurrently, the decline was more marked (to 56.22% and 56.1% resp.), and as compared with the calcitonin treated group (to 67.73% and 82.30% resp.); the difference is significant at the 5% level. The decline of UPD and UDPD values persisted also three months after discontinued treatment; in patients on the diet with collagen hydrolysate practically no rise of these indicators occurred (54.02% and 56.66% resp.).

Conclusions: a) administration of 100 u. calcitonin twice a week for 24 weeks led to a decline of excretion indicators of bone collagen breakdown products, b) the effect of treatment must be monitored using these indicators, c) oral administration of collagen proteins enhanced and prolonged the effect of calcitonin.

6th.Abstract

Specific Collagen Peptides Improve Bone Mineral Density and Bone Markers in Postmenopausal Women—A Randomized Controlled Study

Daniel König, Stephan Scharla, Denise Zdzieblik, Steffen Oesser, Albert Golhofer

Introduction:

Investigations in rodents as well as in vitro experiments have suggested an anabolic influence of specific collagen peptides (SCP) on bone formation and bone mineral density (BMD). The goal of the study was to investigate the effect of 12-month daily oral administration of 5 g SCP vs. placebo (CG: control group) on BMD in postmenopausal women with primary, age-related reduction in BMD.

Methods:

131 women were enrolled in this randomized, placebo-controlled double-blinded investigation. The primary endpoint was the change in BMD of the femoral neck and the spine after 12 months. In addition, plasma levels of bone markers—amino-terminal propeptide of type I collagen (PINP) and C-telopeptide of type I collagen (CTX 1)—were analysed.

Results:

A total of 102 women completed the study, but all subjects were included in the intention-to-treat (ITT) analysis (age 64.3 ± 7.2 years; Body Mass Index, BMI 23.6 ± 3.6 kg/m²; T-score spine -2.4 ± 0.6 ; T-score femoral neck -1.4 ± 0.5). In the SCP group (n= 66), BMD of the spine and of the femoral neck increased significantly compared to the control group (n= 65) (T-score spine: SCP $+0. \pm 0.26$; CG $- 0.03 \pm 0.18$; ANCOVA p= 0.030; T-score femoral neck: SCP $+0.09 \pm 0.24$; CG $- 0.01 \pm 0.19$; ANCOVA p= 0.003). PINP increased significantly in the SCP group (p= 0.007), whereas CTX 1 increased significantly in the control group (p= 0.011).

Conclusions:

These data demonstrate that the intake of SCP increased BMD in postmenopausal women with primary, age-related reduction of BMD. In addition, SCP supplementation was associated with a favorable shift in bone markers, indicating increased bone formation and reduced bone degradation.

7th.Abstract

A nutritional intervention study with hydrolyzed collagen in pre-pubertal spanish children: influence on bone modeling biomarkers

Elena Martin-Bautista 1, Miguel Martin-Matillas, Jose A Martin-Lagos, M Teresa Miranda-Leon, Manuel Muñoz-Torres, Estrella Ruiz-Requena, M Rivero, Josefa Quer, Isabel Puigdueta, Cristina Campoy

Aim: The aim of the study was to investigate the influence of dietary intake of commercial hydrolyzed collagen (Gelatine Royal) on bone remodeling in pre-pubertal children.

Methods: A randomized double-blind study was carried out in 60 children (9.42 +/- 1.31 years) divided into three groups according to the amount of partially hydrolyzed collagen taken daily for 4 months: placebo (G-I, n=18), collagen (G-II, n=20) and collagen+calcium (G-III, n=22) groups. Analyses of the following biochemical markers were carried out: total and bone alkaline phosphatase (tALP and bALP), osteocalcin, tartrate-resistant acid phosphatase (TRAP), type I collagen carboxy-terminal telopeptide, lipids, calcium, 25-hydroxyvitamin D, insulin-like growth factor-1 (IGF-1), thyroid-stimulating hormone, free thyroxin and intact parathormone.

Results: There was a significantly greater increase in serum IGF-1 in G-III than in G-II ($p < 0.01$) or G-I ($p < 0.05$) during the study period, and a significantly greater increase in plasma tALP in G-III than in G-I ($p < 0.05$). Serum bALP behavior significantly ($p < 0.05$) differed between G-II (increase) and G-I (decrease). Plasma TRAP behavior significantly differed between G-II and G-I ($p < 0.01$) and between G-III and G-II ($p < 0.05$).

Conclusion: Daily dietary intake of hydrolyzed collagen seems to have a potential role in enhancing bone remodeling at key stages of growth and development.

8th.Abstract

Association of collagen type I alpha 1 +1245G/T polymorphism and osteoporosis risk in post-menopausal women: a meta-analysis

Jian Wu , Meng Yu , Yixin Zhou

Aim: To fully understand the association between collagen type I alpha 1 (COL1A1) +1245G/T polymorphism and osteoporosis risk in post-menopausal women.

Methods: We searched the electronic databases including PubMed, Embase and Cochrane library updated to January 2016, as well as printed articles. Studies were screened according to the predefined inclusion and exclusion criteria, and the included studies were further assessed by Clark scores system. Furthermore, the association between COL1A1 +1245G/T polymorphism and osteoporosis risk was assessed using odds ratios (ORs) and their 95% confidence intervals (95% CIs). Additionally, the meta-analysis was carried out using Review Manger 5.1 software (Cochrane Collaboration, Oxford, UK) and Stata 11.0 software (StataCorp, College Station, TX, USA).

Results: A total of five studies, which contained 1557 subjects (including 458 osteoporosis patients, 269 osteopenic patients and 830 normal controls), were included. Further, the included studies achieved Clark scores no less than 5, indicating a moderate or high quality. The pooled effect size indicated that there was a significant association between COL1A1 +1245G/T polymorphism and osteoporosis risk only under a co-dominant model (GG vs. TT, OR = 2.45, 95% CI = 1.11-5.44, P = 0.03). Moreover, there was no significant correlation between COL1A1 +1245G/T polymorphism and osteopenic risk under each model (P > 0.05).

Conclusion: There might be a certain relationship between GG genotype of COL1A1 +1245G/T polymorphism and osteoporosis risk in post-menopausal women.

9th Abstract

Osteoporosis Changes Collagen/Apatite Orientation and Young's Modulus in Vertebral Cortical Bone of Rat

Ryosuke Ozasa, Takuya Ishimoto, Sayaka Miyabe, Jun Hashimoto, Makoto Hirao, Hideki Yoshikawa & Takayoshi Nakano

This study revealed the distinguished changes of preferential orientation of collagen and apatite and Young's modulus in two different types of osteoporotic bones compared with the normal bone. Little is known about the bone material properties of osteoporotic bones; therefore, we aimed to assess material properties in osteoporotic bones. 66 female Sprague–Dawley rats were used. We analyzed the volumetric bone mineral density, collagen/apatite orientation, and Young's modulus of fifth lumbar vertebral cortex for osteoporotic rats caused by ovariectomy (OVX), administration of low calcium and phosphate content (LCaP) diet, and their combination (OVX + LCaP), as well as sham-operated control. Osteocyte conditions were assessed by hematoxylin and eosin and immunohistochemical (matrix extracellular phosphoglycoprotein (MEPE) and dentin matrix protein 1 (DMP1)) staining. All osteoporotic animals showed bone loss compared with the sham-operated control. OVX improved craniocaudal Young's modulus by enhancing collagen/apatite orientation along the craniocaudal axis, likely in response to the elevated stress due to osteoporotic bone loss. Conversely, LCaP-fed animals showed either significant bone loss or degraded collagen/apatite orientation and Young's modulus. Osteocytes in LCaP and OVX + LCaP groups showed atypical appearance and MEPE- and DMP1-negative phenotype, whereas those in the OVX group showed similarity with osteocytes in the control group. This suggests that osteocytes are possibly involved in the osteoporotic changes in collagen/apatite orientation and Young's modulus. This study is the first to demonstrate that osteoporosis changes collagen/apatite orientation and Young's modulus in an opposite manner depending on the cause of osteoporosis in spite of common bone loss.

10th Abstract

A double-blind, placebo-controlled, randomised, clinical study on the effectiveness of collagen peptide on osteoarthritis

Suresh Kumar, Fumihito Sugihara, Keiji Suzuki, Naoki Inoue, Sriraam Venkateswarathirukumara

Background: Recent studies show that enzymatically hydrolysed collagen, the collagen peptide, is absorbed and distributed to joint tissues and has analgesic and anti-inflammatory properties. A double-blind, placebo-controlled, randomised trial with collagen peptides isolated from pork skin (PCP) and bovine bone (BCP) sources was carried out to study the effectiveness of orally supplemented collagen peptide to control the progression of osteoarthritis in patients diagnosed with knee osteoarthritis. Improvement in treatment was assessed with reduction in Western Ontario McMaster Universities (WOMAC), visual analogue scale (VAS) and quality of life (QOL) scores from baseline to 13 weeks (Visit 7). Safety and tolerability were also evaluated.

Results: There was significant reduction from baseline to Visit 7 in the primary end points of WOMAC and VAS scores and in the secondary end point of QOL score in subjects with PCP and BCP groups, while in subjects with placebo group the end point indices remained unaltered. Furthermore, all the score levels of WOMAC, VAS and QOL decreased significantly ($P < 0.01$) in the study group compared to placebo group in Visit 7.

Conclusion: The study demonstrated that collagen peptides are potential therapeutic agents as nutritional supplements for the management of osteoarthritis and maintenance of joint health.

11 th Abstract

The effects of collagen peptides on muscle damage, inflammation and bone turnover following exercise: a randomized, controlled trial

Tom Clifford, Matthew Ventress, Dean M Allerton, Sarah Stansfield, Jonathan C Y Tang, William D Fraser, Barbara Vanhoecke, Janne Prawitt, Emma Stevenson

This study examined whether consuming collagen peptides (CP) before and after strenuous exercise alters markers of muscle damage, inflammation and bone turnover. Using a double-blind, independent group's design, 24 recreationally active males consumed either 20 g day⁻¹ of CP or a placebo control (CON) for 7 days before and 2 days after performing 150 drop jumps. Maximal isometric voluntary contractions, countermovement jumps (CMJ), muscle soreness (200 mm visual analogue scale), pressure pain threshold, Brief Assessment of Mood Adapted (BAM+) and a range of blood markers associated with muscle damage, inflammation and bone turnover C-terminal telopeptide of type 1 collagen (β -CTX) and N-terminal propeptides of type 1 pro-collagen (PINP) were measured before supplementation (baseline; BL), pre, post, 1.5, 24 and 48 h post-exercise. Muscle soreness was not significantly different in CP and CON ($P = 0.071$) but a large effect size was evident at 48 h post-exercise, indicative of lower soreness in the CP group (90.42 ± 45.33 mm vs. CON 125.67 ± 36.50 mm; ES = 2.64). CMJ height recovered quicker with CP than CON at 48 h ($P = 0.050$; CP 89.96 ± 12.85 vs. CON $78.67 \pm 14.41\%$ of baseline values; ES = 0.55). There were no statistically significant effects for the other dependent variables ($P > 0.05$). β -CTX and PINP were unaffected by CP supplementation ($P > 0.05$).

In conclusion, CP had moderate benefits for the recovery of CMJ and muscle soreness but had no influence on inflammation and bone collagen synthesis.